

Prop sistema ortogonal indep lineal

Proposición 1. Sea H un espacio prehilbert y sea S un sistema ortogonal

$\implies S$ es linealmente independiente.

Demostración: Sean $\{\alpha_i\}_{i \in \mathbb{N}_n} \subset \mathbb{K}$ y $\{x_i\}_{i \in \mathbb{N}_n} \subset S$ tales que $\sum_{i=1}^n \alpha_i x_i = 0$. Entonces,

$$0 = \left\| \sum_{i=1}^n \alpha_i x_i \right\|^2 \stackrel{\text{Pitágoras}}{=} \sum_{i \in \mathbb{N}_n} \|\alpha_i x_i\|^2 = \sum_{i=1}^n |\alpha_i|^2 \|x_i\|^2.$$

Como $\forall i \in \mathbb{N}_n : \|x_i\|^2 > 0$, $\forall i \in \mathbb{N}_n : \alpha_i = 0$, luego S es linealmente independiente. ■

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